

Layered No-Refolding Witnesses in the Hilbert Substrate Framework

Ben Bray

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Abstract

This note upgrades the current no-refolding program from graph-level surrogates to a layered witness framework. The main claim is that no-refolding should not be witnessed by generic graph persistence or shell-size preservation alone. Instead, it should be witnessed by failure to preserve committed bookkeeping structure. The proposed witness stack has four layers: (1) interface bookkeeping preservation, (2) slack creation, (3) collective relief-function preservation, and (4) sector-specific bookkeeping preservation. The result is a general no-refolding witness with sector-specific specializations for fermionic, gauge/interface, and mesoscopic collective-mode sectors.

1 Purpose

The current HSF picture already suggests that:

- real links are interface sectors rather than mere graph edges,
- a working tensor split is not freely mutable once it is carrying real structure,
- no-signaling alone is too weak because slack-heavy solutions survive,
- finite bandwidth should be read through active influence channels rather than occupancy counts alone.

The problem is that existing mesoscopic witnesses have often been too graphy: shell size, triangle count, core score, edge persistence, or spectral geometry. These are useful diagnostics, but they do not yet directly tell us whether a move preserved or destroyed the committed bookkeeping that a working collective structure depends on.

This note proposes a layered witness for no-refolding proper.

2 Guiding principle

The intended principle is:

A move is refolding-like when it destroys committed bookkeeping without preserving the same collective relief function elsewhere.

That means no-refolding should be witnessed by more than one type of failure. At minimum, a refolding-like move should:

1. disrupt interface bookkeeping,

2. or create new slack / dead directions,
3. or destroy the relief function of the existing object,
4. and in sector-specific cases, alter protected bookkeeping for that sector.

Thus the witness is layered rather than singular at first.

3 General layered witness

For a candidate move M in a local region R , define the general no-refolding witness

$$W_{\text{NR}}(M) = \alpha_{\text{ep}} W_{\text{ep}}(M) + \alpha_{\text{slack}} W_{\text{slack}}(M) + \alpha_{\text{func}} W_{\text{func}}(M), \quad (1)$$

with nonnegative coefficients

$$\alpha_{\text{ep}}, \alpha_{\text{slack}}, \alpha_{\text{func}} \geq 0.$$

Interpretation:

- W_{ep} : loss of endpoint/interface bookkeeping,
- W_{slack} : increase in underused dead directions,
- W_{func} : loss of collective relief function.

A move is more refolding-like when W_{NR} is larger.

4 Layer 1: interface bookkeeping preservation

4.1 Motivation

A real HSF link is not just an adjacency relation; it is an interface register with independent left/right endpoint structure. If no-refolding is what makes that interface committed, then a move should be penalized when it changes which endpoint structure is actually doing the work.

4.2 Endpoint-structure loss witness

Let E_R^{comm} be the set of committed links in region R . For each $e \in E_R^{\text{comm}}$, let

$$I_e^L, I_e^R$$

be the left and right endpoint influence maps before the move, and let

$$I_{e,M}^L, I_{e,M}^R$$

be the corresponding maps after applying move M .

Define the endpoint-structure preservation score

$$P_{\text{ep}}(M) = \frac{1}{|E_R^{\text{comm}}|} \sum_{e \in E_R^{\text{comm}}} \frac{\langle I_e^L, I_{e,M}^L \rangle_F + \langle I_e^R, I_{e,M}^R \rangle_F}{\|I_e^L\|_F \|I_{e,M}^L\|_F + \|I_e^R\|_F \|I_{e,M}^R\|_F + \varepsilon}, \quad (2)$$

where $\langle A, B \rangle_F = \text{tr}(A^\dagger B)$ is the Frobenius inner product.

Then define

$$W_{\text{ep}}(M) = 1 - P_{\text{ep}}(M). \quad (3)$$

Interpretation:

- $W_{\text{ep}} \approx 0$: same interface bookkeeping is still active,
- large W_{ep} : committed endpoint structure has been rewritten.

5 Layer 2: slack creation

5.1 Motivation

No-signaling alone is too weak because reducible or slack-heavy links can still satisfy endpoint commutativity while carrying many underused dead directions. Thus no-refolding should penalize moves that inflate the same interface into a looser, more padded, more reducible structure.

5.2 Slack witness

For each committed link e , let r_e be the effective active-channel rank, for example derived from the singular values of an influence map, and let $d_{\text{act}}(e)$ be the active local interface dimension available to the link. Define

$$\text{slack}(e) = d_{\text{act}}(e) - r_e. \quad (4)$$

After move M , define the moved slack $\text{slack}_M(e)$. Then the move-induced slack creation is

$$W_{\text{slack}}(M) = \frac{1}{|E_R^{\text{comm}}|} \sum_{e \in E_R^{\text{comm}}} [\text{slack}_M(e) - \text{slack}(e)]_+, \quad (5)$$

where $[x]_+ = \max(x, 0)$.

Interpretation:

- small W_{slack} : the move did not create underused dead directions,
- large W_{slack} : the move made the same interface bookkeeping looser and more reducible.

6 Layer 3: collective relief-function preservation

6.1 Motivation

No-refolding is not only about preserving endpoint algebra. It is also about preserving the collective function of the object that was carrying local constraint stress. A move that changes support but preserves the same relief function elsewhere should count as lawful re-expression, not refolding.

6.2 Function-loss witness

Define

$$W_{\text{func}}(M) = \beta_F(1 - F_{\text{org}}(M)) + \beta_I[\Delta I_{\text{lost}}(M)]_+ + \beta_A(1 - R_{\text{act}}(M)), \quad (6)$$

where:

- $F_{\text{org}}(M) \in [0, 1]$ is organizer fidelity between baseline and moved branches,
- $\Delta I_{\text{lost}}(M)$ is local mutual-information loss under the move,
- $R_{\text{act}}(M) \in [0, 1]$ is retained committed-interface activity.

The retained activity may be defined, for example, as

$$R_{\text{act}}(M) = \frac{\sum_{e \in E_R^{\text{comm, after}}} A_e^{\text{after}}}{\sum_{e \in E_R^{\text{comm, before}}} A_e^{\text{before}} + \varepsilon}, \quad (7)$$

where A_e is any accepted link-activity score (such as bounded summed singular values).

Interpretation:

- small W_{func} : the same relief function is still being carried,
- large W_{func} : the object's function was actually degraded.

7 Sector-specific bookkeeping witnesses

The general witness W_{NR} is not yet complete in sectors where additional bookkeeping must be preserved.

7.1 Fermionic sector: Jordan–Wigner bookkeeping

If fermionic structure is present, then the move should also preserve the nonlocal sign/order bookkeeping required for the same excitation algebra.

Let $\mathcal{J}_R$ be the effective Jordan–Wigner ordering/string map in region R , and $\mathcal{J}_{R,M}$ the post-move map. Let P_R be the set of physically occupied or active fermionic excitation supports in that region.

Define a hard mismatch witness

$$W_{\text{JW}}(M) = \frac{1}{|P_R|} \sum_{p \in P_R} \mathbf{1}[\chi_p(\mathcal{J}_R) \neq \chi_p(\mathcal{J}_{R,M})], \quad (8)$$

where χ_p is the parity/sign string seen by excitation p .

A softer overlap form is

$$W_{\text{JW}}^{\text{soft}}(M) = 1 - \frac{1}{|P_R|} \sum_{p \in P_R} \text{overlap}(\chi_p(\mathcal{J}_R), \chi_p(\mathcal{J}_{R,M})). \quad (9)$$

Interpretation:

- small W_{JW} : the move preserved fermionic bookkeeping,
- large W_{JW} : support changed while effective JW/sign-order structure changed as if it were free relabeling.

7.2 Gauge/interface sector: compatibility preservation

If the sector carries gauge-compatible interface structure, the move should also preserve local attachment compatibility.

Let Γ_e^{before} and Γ_e^{after} be the local compatibility descriptors for committed link e before and after the move. Then define

$$W_{\text{gauge}}(M) = \frac{1}{|E_R^{\text{comm}}|} \sum_{e \in E_R^{\text{comm}}} \|\Gamma_e^{\text{before}} - \Gamma_e^{\text{after}}\|. \quad (10)$$

Interpretation:

- small W_{gauge} : local compatibility was preserved,
- large W_{gauge} : the move rewrote gauge/interface bookkeeping.

7.3 Mesoscopic object sector: collective-mode preservation

At the mesoscopic level, your organizer program already suggests that objects have a response hierarchy under collective perturbations. Thus a move should also be judged by whether it preserves the object's characteristic response structure.

Let

$$\mathbf{r}_{\text{before}} = (r_{\text{dip}}, r_{\text{breath}}, r_{\text{circ}}, \dots)_{\text{before}}$$

be the standardized response vector of the object to a fixed panel of small collective probes, and likewise $\mathbf{r}_{\text{after}}$ after the move.

Define

$$W_{\text{mode}}(M) = 1 - \text{overlap}(\mathbf{r}_{\text{before}}, \mathbf{r}_{\text{after}}). \quad (11)$$

Interpretation:

- small W_{mode} : the same collective object still exists in functional form,
- large W_{mode} : the move effectively rewrote the object's response identity.

8 Full layered witnesses

The general no-refolding witness is:

$$W_{\text{NR}}(M) = \alpha_{\text{ep}} W_{\text{ep}}(M) + \alpha_{\text{slack}} W_{\text{slack}}(M) + \alpha_{\text{func}} W_{\text{func}}(M) \quad (12)$$

The fermionic specialization is:

$$W_{\text{ferm}}(M) = W_{\text{NR}}(M) + \alpha_{\text{JW}} W_{\text{JW}}(M) \quad (13)$$

The gauge/interface specialization is:

$$W_{\text{gauge}}^{\text{full}}(M) = W_{\text{NR}}(M) + \alpha_{\Gamma} W_{\text{gauge}}(M) \quad (14)$$

The mesoscopic object specialization is:

$$W_{\text{obj}}(M) = W_{\text{NR}}(M) + \alpha_{\text{mode}} W_{\text{mode}}(M) \quad (15)$$

Thus the general witness is framework-level, and the extra bookkeeping witness is added only when that sector is relevant.

9 Decision status

Lawful enough now

The following witness components are strong enough to use now as provisional lawful witnesses:

- W_{func} : organizer fidelity + local MI loss + retained activity,
- W_{slack} : slack increase from active-channel rank / underused directions,
- W_{ep} , provided explicit endpoint influence maps are available.

Needs concrete formalization

The following remain underived operationally:

- exact choice of endpoint map and overlap in W_{ep} ,
- exact effective JW/string map in W_{JW} ,
- exact local compatibility descriptor in W_{gauge} ,
- exact standardized probe panel in W_{mode} .

10 Working interpretation

Under this framework:

- a move counts as lawful re-expression if it changes support while keeping W_{NR} small,
- a move counts as unlawful refolding if it drives W_{NR} large,
- and in a specific sector, sector bookkeeping failure must also remain small.

Thus no-refolding is no longer just “do not destroy a good graph motif.” It is:

Do not destroy committed interface bookkeeping and collective relief function, and in sectors with extra bookkeeping requirements, do not destroy those either.

11 Conclusion

The main result of this note is the following layered statement:

No-refolding proper should be witnessed by failure to preserve committed interface bookkeeping, by creation of new slack/dead directions, and by loss of collective relief function. Sector-specific bookkeeping—such as fermionic Jordan–Wigner sign/order structure, local gauge/interface compatibility, or mesoscopic collective-mode identity—should then be added as an extra witness layer when that sector is physically relevant.

This is a stronger and more disciplined witness framework than shell size, edge count, or graph persistence alone. It gives a path toward an actual no-refolding term in a lawful churn rule.

One-sentence summary

The upgraded no-refolding witness is:

interface bookkeeping loss + slack creation + relief-function loss + sector-specific bookkeeping loss